

Forecasting Market Volatility (VIX) with Federal Reserve Language

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Introduction

Goal: Train a DL model on Federal Reserve text to identify language patterns predictive of the 10-day forward change in the VIX

The Federal Reserve (Fed): U.S. central bank

- Sets interest rates and monetary policy in financial markets
- Independent of politics, but Chair is presidentially appointed

The VIX: S&P 500's 30-day implied volatility

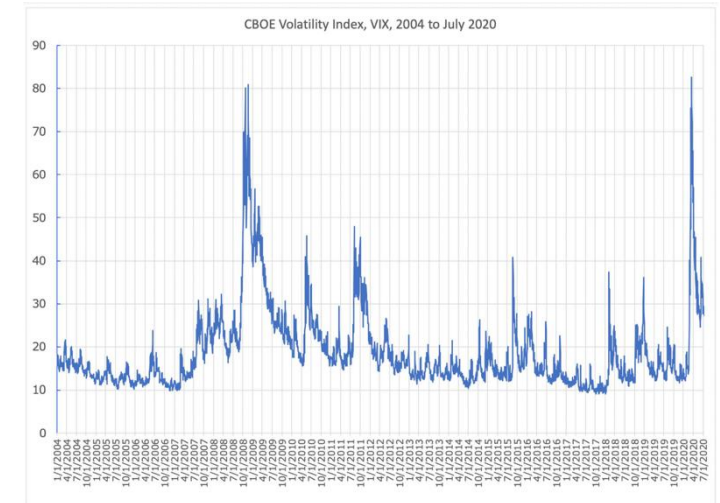
- widely known as the “fear gauge”

Personal Interest: how politics affect the stock market?

Difference in the Trump vs Biden regimes?

Hypothesis: when Fed text is surprising or hawkish, expected volatility moves

Prior work shows Fed text moves Treasury yields and equities, limited research on VIX



Dataset: 326 Fed documents (1993-2026)

Sources from [federalreserve.gov](https://www.federalreserve.gov) (5-10k words per doc)

- Federal Open Market Committee (FOMC) minutes (268)
 - committee that meets 8 times a year and decides monetary policy
- Humphrey-Hawkins testimony (58)
 - twice-yearly testimony Fed Chair reports to Congress on monetary policy

Preprocessing

- First 80 sentences to best capture the policy rationale

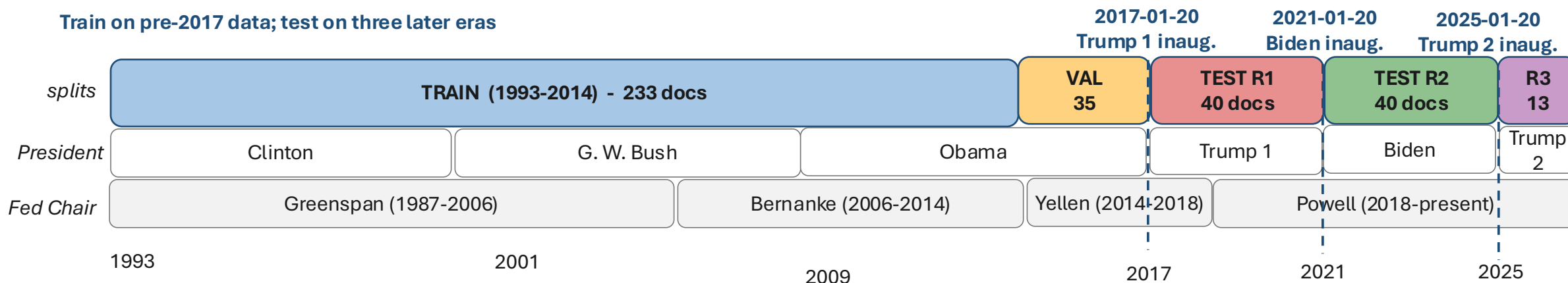
Target: 10-day forward change in CBOE VIX (FRED VIXCLS)

- Aligned to next trading day to avoid look-ahead leakage

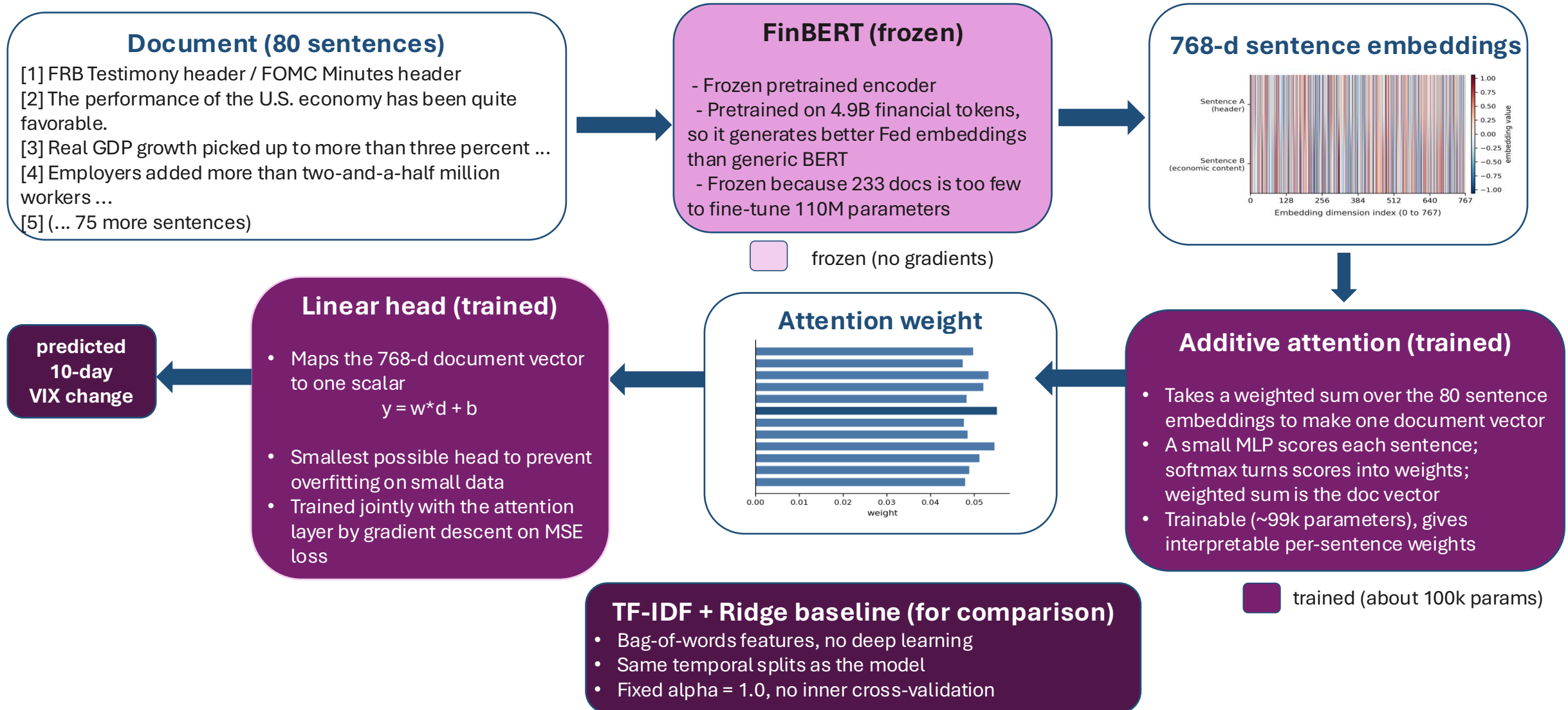
- [1] FRB: Testimony, Greenspan - Monetary Policy Report to the Congress, U.S. Senate --July 20, 2004 Testimony of Chairman Alan Greenspan Federal Reserve Board's semiannual Monetary Policy Report to the Congress Befo...
- [2] Economic developments in the United States have generally been quite favorable in 2004, lending increasing support to the view that the expansion is self-sustaining.
- [3] Not only has economic activity quickened, but the expansion has become more broad-based and has produced notable gains in employment.
- [4] The evident strengthening in demand that underlies this improved performance doubtless has been a factor contributing to the rise in inflation this year.
- [5] But inflation also seems to have been boosted by transitory factors such as the surge in energy prices.

Example Humphrey-Hawkins testimony 2004-07-20

Train on pre-2017 data; test on three later eras



FinBERT + Sentence Attention + Linear Head



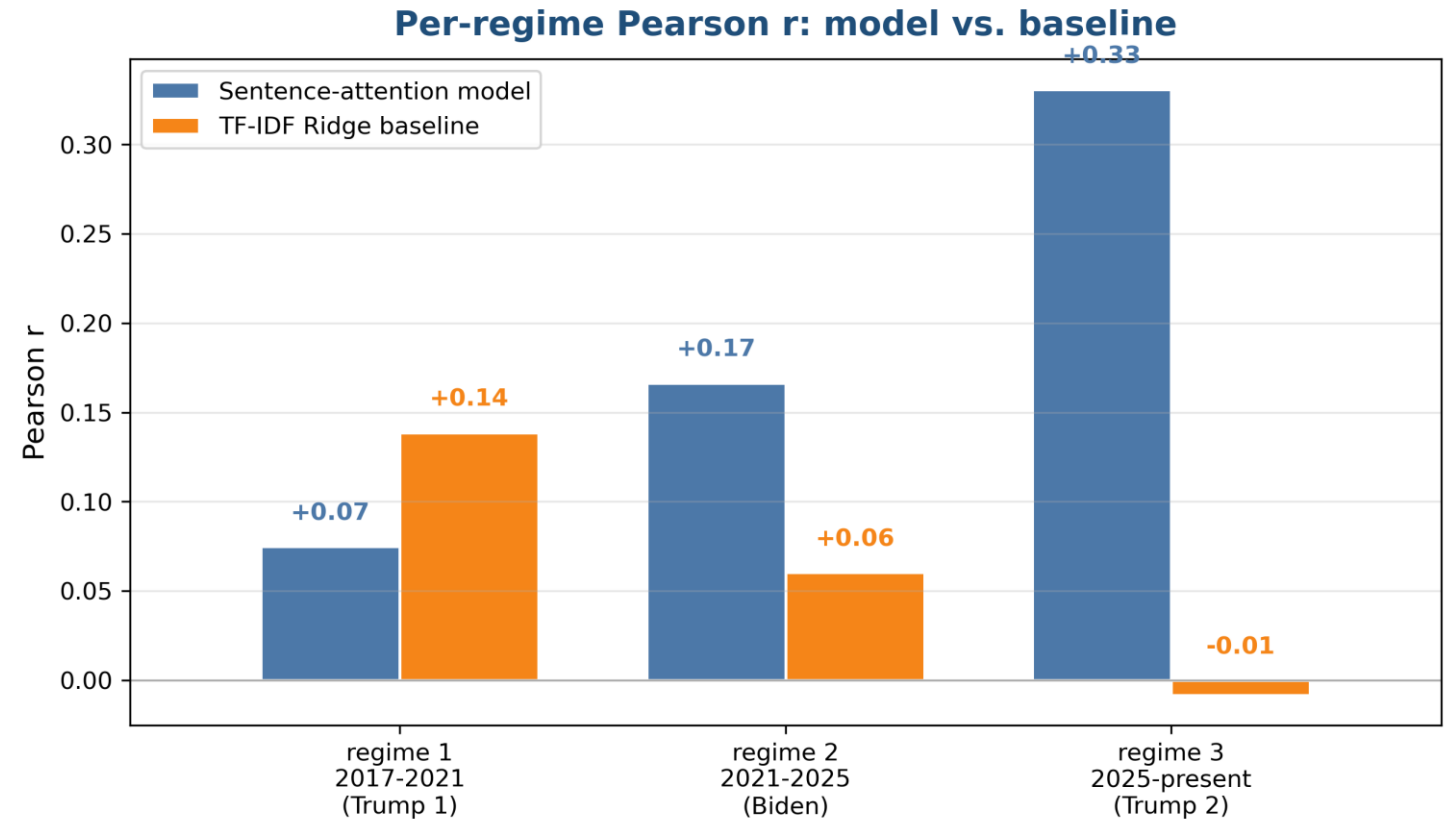
Results

Evaluation Metric

- Pearson r and p-value
- Predicted-vs-actual correlation (r) and probability it's just noise (p)

Results

- Validation set: $r = +0.34$, $p = 0.045$
- Model holds up across three later eras it never saw in training
- Beats TF-IDF Ridge baseline in regime 2 (Biden) and regime 3 (Trump 2)
- Regime 3 shows strongest signal: model $r = +0.33$ vs baseline $r = -0.01$



- Caveat: regime 3 has only $n = 13$ documents, whereas 1 and 2 have 40
- Wider confidence interval
 - regime 3's strong score could partly be noise from the small sample

Conclusions

- What worked
 - Biggest finding: target horizon mattered more than the architecture.
 - Switching from 3-day to 10-day moved the result from null to statistically significant
 - Using frozen FinBERT for pretrained financial embeddings on small data
 - Attention pooling gave interpretable weights
- Anything you would do differently
 - FOMC statements (same-day release) instead of minutes (3-week delay)
 - Intraday VIX target to capture the immediate market reaction
 - More data, especially to firm up the recent regimes